

YASSIEN TAWFIK

BIOMEDICAL AI ENGINEER

 Cairo, Egypt

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PROFILE

Biomedical engineer working on deep learning for brain MRI segmentation where annotated data is scarce, with a bias toward models that generalize across scanners rather than ones tuned to a single dataset. Starting a research MSc at the University of Calgary in August 2026.

EDUCATION

M.Sc. Biomedical Engineering — University of Calgary, Canada **Aug 2026 – 2028**

Research-based program at the AI²Lab under Dr. Roberto Souza.

Focus: AI for neonatal brain MRI.

B.Sc. Biomedical Engineering — Cairo University, Egypt | GPA: 3.4/4.0 **Sep 2021 – 2026**

GRADUATION PROJECT

Automated FCD Lesion Segmentation for Pre-Surgical Epilepsy Evaluation | [\[GitHub Link\]](#)

Supervisors: Dr. Aya Khalaf (Yale), Dr. Tamer Basha, Eng. Mahmoud Salman

Two segmentation architectures released as a single submodule-based suite.

- **nnU-FCD** — nnU-Net benchmarked against baseline U-Net variants for FCD II lesion segmentation on FLAIR MRI; multimodal preprocessing and cross-sequence evaluation
- **SynthFCD** — SynthSeg-inspired synthetic data generation with custom lesion synthesis and 3D domain-randomized training; cuts reliance on scarce annotations and holds up across scanners.

PUBLICATIONS

Y. Tawfik, M. Marwan, M. Yasser, N. Mahmoud, M. Mosaad, M. Salman, T. Basha, and A. Khalaf, "Evaluation of nnU-Net for FCD II Lesion Segmentation in FLAIR MRI," in *Proc. 48th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Toronto, Canada, Jul. 2026, in press.

Oral presentation · Student Paper Competition finalist

TECHNICAL SKILLS

- **AI & Deep Learning** — 3D medical image segmentation (nnU-Net, U-Net); synthetic data generation & domain randomization; cross-scanner generalization; explainability (Grad-CAM, SHAP)
- **Neuroimaging & Medical Imaging** — MRI/CT preprocessing; FLAIR lesion segmentation; NIfTI handling, intensity normalization, registration; Dice-based evaluation & k-fold cross-validation
- **ML Engineering** — PyTorch, MONAI, TensorFlow, scikit-learn; GPU training at scale (mixed precision, checkpointing, CUDA debugging); modular pipeline design; Git submodule repo architecture
- **Programming & Tools** — Python, C/C++; NumPy, Pandas, SciPy, OpenCV; Git; Linux; cloud GPU environments
- **Biomedical Domain** — Biomedical signal processing (ECG filtering & feature extraction); CT/MRI hardware fundamentals; clinical imaging workflows

PROFESSIONAL EXPERIENCE

Part-time Product Specialist | Optoscient

Aug 2025 - Jun 2026

Leading Egyptian distributor; digital pathology solutions; AI-integrated software platforms; technical support for pathology systems and diagnostic software; clinical team assistance; imaging hardware implementation and integration; cloud-based analysis tools; workflow efficiency; diagnostic accuracy enhancement.

INTERNSHIPS & COURSES

AI Developer Intern — Elevvo.tech 

Clinical Engineer Intern — BAHEYA Foundation

MRI & CMR Course — Siemens Healthineers

CT Essentials Course — Siemens Healthineers

PROJECTS

• **Brain MRI Tumor Analysis Platform**

- AI-powered web platform; Brain tumor classification; automated reporting; API integration; Xception model; web deployment; AI integration; clinical workflow. [\[GitHub Link\]](#)

• **MRI K-Space Reconstruction and Frequency Analysis Tool**

- *K-space manipulations using FFT-based filtering; demonstrates undersampling and artifact formation; evaluates reconstruction quality via quantitative metrics.* [\[GitHub Link\]](#)

• **Explainable Pediatric Chest Radiograph Diagnosis**

- *Fine-tuned ResNet-50; two-stage transfer learning; Grad-CAM explainability; pediatric pneumonia detection; 93.6% sensitivity; clinical interpretability.* [\[GitHub Link\]](#)

• **Explainable Breast Cancer Classification**

- *XAI, SHAP; SMOTE balancing; binary tumor classification; ROC-AUC ~99%; explainability; imbalanced data handling; clinical metrics.* [\[GitHub Link\]](#)

• **Oral Cancer Risk Prediction**

- *Random Forest; microbiome analysis; TCMA dataset; feature selection & SHAP explainability; accuracy 92.89%, AUROC 0.97; ML; non-invasive diagnostics.* [\[GitHub Link\]](#)

• **Dual-Energy X-Ray Material Decomposition System**

- *Dual-energy acquisition simulation; mathematical material decomposition; bone and soft-tissue separation; quantitative imaging; medical physics modeling.* [\[GitHub Link\]](#)

• **Advanced Fourier Image Analysis Platform**

- *FFT-based imaging; frequency-domain filtering; region masking; Gaussian smoothing; multi-image Fourier mixing; real-time image reconstruction workflows.* [\[GitHub Link\]](#)

• **Python Imaging and Vision Toolkit**

- *Image processing & CV; edge detection; segmentation; feature extraction; modular Python toolkit; software architecture; reusable CV modules.* [\[GitHub Link\]](#)

• **Real-Time ECG Arrhythmia Detection**

- *Deep learning (CNN); ECG visualization; arrhythmia detection; TensorFlow; real-time signal processing; CNN for biomedical data; GUI integration.* [\[GitHub Link\]](#)

• **Physics-Based Doppler Ultrasound Simulator**

- *Medical-grade physics engine; real-time STFT spectral analysis; laminar flow modeling; hemodynamic simulation; signal processing workflow.* [\[GitHub Link\]](#)

• **ECG-Guided Automated Defibrillation**

- *Hardware/software integration; real-time ECG monitoring; tachycardia detection; GUI visualization; embedded systems + biomedical signal workflow.* [\[GitHub Link\]](#)

• **Multi-Agent LLM Decision Support System**

- *Sequential three-agent Gemini pipeline with synthesizer aggregation; custom Python orchestrator; session-bound BYOK key handling; Streamlit deployment (live).* [\[GitHub Link\]](#)